

Derbyshire Shared Care Pathology Guidelines

First Line Investigation of Suspected Hyperaldosteronism for Primary Care

Primary aldosteronism (PA) involves about 5% of the hypertensives seen by general practitioners, more than 11% of those referred to specialized hypertension centres and up to 20% of those with difficult to treat hypertension.

The main causes of primary hyperaldosteronism are: bilateral idiopathic adrenal hyperplasia (incidence ~65%), unilateral aldosterone-producing adenoma (Conn's Syndrome) (incidence ~30%), unilateral primary adrenal hyperplasia (incidence ~3%), Glucocorticoid-suppressible hyperaldosteronism (incidence ~1%), Adrenocortical carcinoma (incidence <1%).

The use of single simultaneous samples for aldosterone and renin measurement, with derivation of the "aldosterone/renin ratio" has been advocated as the first line in the investigation of suspected hyperaldosteronism. Low or undetectable renin levels along with inappropriately raised plasma aldosterone concentration (PAC) are the biochemical hallmarks of PA.

Since the renin-aldosterone axis is primarily regulated by renal blood flow, subjects under investigation should not be taking any drugs that interfere with fluid balance or potassium. Subjects should also be adequately hydrated and have an adequate oral intake of sodium. Hypokalaemia should be avoided since it suppresses aldosterone secretion (see under patient preparation for details).

INDICATIONS

This test is indicated in the differential diagnosis of secondary hypertension.

- Hypertension and hypokalaemia (spontaneous or diuretic induced)
- Resistant hypertension (≥ 150 mm Hg resistant to 3 conventional antihypertensive drugs)
- Adrenal "incidentaloma" ≥ 10 mm and hypertension
- Hypertension with a Family history of primary hyperaldosteronism
- Hypertension onset or Stroke before 40 years of age

PREPARATION

Patient

- Give potassium replacement (Slow K or Sando-K tabs) sufficient to raise serum potassium into reference range (3.5 – 5.3 mmol/L)
- Patients should be normally hydrated and have an adequate oral intake of sodium
- Patients should be ambulatory for at least 120 minutes before the test (performing usual activities), however samples are best taken between 08:00 and 10:00 during the diurnal peak of aldosterone secretion and after 5-15 mins of sitting

Stopping medications: The table below indicates drugs which can interfere with renin and/or aldosterone measurements. Most anti-hypertensives are implicated but α -adrenoceptor blockers (e.g. Doxazosin), slow release Verapamil and Hydralazine are thought to have minimal effects.

Ideally all drug therapy should be stopped at least 2-3 weeks before testing, but this is not generally practical. The test does not give useful results in patients treated with spironalactone or eplerenone.

A pragmatic approach is to stop spironalactone and eplerenone in order to send initial aldosterone:renin ratio samples, then stop all necessary medication in order to perform the test under ideal conditions. If the initial results are suggestive of primary hyperaldosteronism then consider secondary confirmatory tests for primary hyperaldosteronism at the same time as repeating the renin:aldosterone ratio under ideal conditions.

When stopping antihypertensives for testing replacement with Doxazosin is often appropriate.

Please be sure to include details of all current drug therapy on the request form.

Drug	Physiological effect	Potential effect on interpretation	Time to remove interference
Spironolactone	Increases renin, variable effect on aldosterone	Uninterpretable	6 weeks
Eplerenone / Amiloride	Increases renin, variable effect on aldosterone	Uninterpretable	2 weeks
Oestrogens	Increase renin substrate	False negative	2 weeks
ACE / AR2B inhibitors	Increase renin and reduce aldosterone	False negative	2 weeks
Beta-blockers	Reduce renin more than aldosterone	False positive	2 weeks
Diuretics	Increase renin and aldosterone	Variable	2 weeks
Hypokalaemia	Inhibits aldosterone secretion	False negative	1 week
NSAIDs	Retain sodium and reduce renin, ?effect on aldosterone	False positive	2 weeks

Equipment

Blood collection tubes: 1 yellow top tube: for U&E
1x 4ml EDTA (purple top): for renin and aldosterone

PROCEDURE

Samples for renin are stable for 6 hours at room temperature.
It is therefore essential that patients are bled within the blood test clinic at the Royal Derby Hospital to ensure the sample is suitable for analysis.

INTERPRETATION

Plasma Renin reference ranges (mIU/L)

Supine	–	<59.7
Ambulant	–	5.3 – 99.1

Plasma Aldosterone reference ranges (pmol/L)

Supine	–	103 - 859
Ambulant	–	103 - 1197

Aldosterone / Renin Ratio (ARR)

Raised ARR (>30 pmol/mIU) suggests possible primary hyperaldosteronism.

Interpretative comments are included on reports.

The aldosterone:renin ratio is only an additional indicator of the state of the aldosterone:renin axis, and must be considered in the light of the actual aldosterone and renin levels. A high aldosterone together with a suppressed renin and elevated aldosterone:renin ratio indicates primary hyperaldosteronism.

Note that:

1. Some patients with significant renal disease may give similar results.
2. For patients taking ACE inhibitors at the time of assessment:
 - detectable renin does not exclude primary hyperaldosteronism
 - undetectable renin would strongly suggest primary hyperaldosteronism

Abnormal results with this test should be followed by further investigation under the supervision of an endocrinologist

ASSAYING LABORATORY

Royal Victoria Infirmary, Newcastle-upon-Tyne

TURNAROUND TIME

Results will normally be available within 3 weeks.

Authors: Biochemistry Standard Clinical Guideline Group, October 1995		
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Dr R Stanworth, Mrs H Seddon	Nov 2015	28th Feb 2020
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